Exchange-Only and Singlet-Only Spin Qubits: Gain-Robust Local Gates, Spin-Sector Detection, and Constraints on Encoded Entanglers

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Abstract

We consolidate a sequence of theoretical and numerical studies of exchange-only (EO) three-spin qubits, four-spin singlet-only exchange-only (SOEO) qubits, and interactions between encoded blocks. The shared control model uses nonnegative isotropic Heisenberg exchange with independent quasi-static fractional gain errors assigned to each physical bond. Within a *strictly serial* control class we report numerical composite-pulse realizations of encoded Hadamard, phase, and Pauli-Z gates whose first-order error generators on every **active** bond are scalar on the encoded space, with vanishing first-order leakage. The duration-optimized Hadamard uses 11 pulses and total normalized exchange area 8.308609π; the selected S and Z constructions use 13 pulses each. Full physical-spin calculations check the EO logical and gauge sectors and the SOEO singlet subspace. Separately, a six-spin collective-exchange and singlet-probe model implements probabilistic detection of departure from the SOEO $S_D = 0$ sector and exhibits first-order cancellation of four independent, static coupling-gain errors at a special revival time. This star construction, however, requires four simultaneous couplings and implements a protected identity, **not** an entangling gate. For two EO blocks we certify an ideal-unitary-preserving 43-to-41-pulse rewrite of a locally rotated controlled phase, with a 16.4% reduction of a specified small-gain response coefficient, while the interblock response remains nonzero. We prove restricted obstructions to single-cross-bond exchange-only entanglers and explain why robust local dressing does not cure a noisy entangling core. This is a research audit, not a claim of hardware performance, fault-tolerant QEC, global gate optimality, or all-bond-robust CZ synthesis.

Keywords: exchange-only qubits; singlet-only encoding; silicon quantum dots; Heisenberg exchange; correlated gain errors; composite pulses; leakage; fault-tolerant syndrome extraction.

1. Scope, status, and relation to prior work

Three-spin EO encodes a qubit in the multiplicity space of total spin $S = 1/2$, with an accompanying $m = \pm 1/2$ gauge degree of freedom. Four-spin SOEO encodes one qubit in the two-dimensional total-spin-zero sector. The recent hybrid EO–Loss–DiVincenzo proposal of Heinz, Sharma and Kattemölle [1] uses *single-spin addressable* ancillas and flagged low-level $XXXX/ZZZZ$ checks. Such ancilla gates are **not native** to a strictly exchange-only controller. We distinguish its physical-spin checks from higher-level checks $X_L^{(A)} X_L^{(B)} X_L^{(C)} X_L^{(D)}$ across four separately encoded EO qubits.

The independent numerical constructions below come from four archived research packages supplied with this report. Earlier dynamically corrected EO gates were developed by Hickman *et al.* [2], and corrected exchange-coupled two-qubit gates were treated by Setiawan *et al.* [3]. Singlet-only always-on gapless (SAGE) qubits [4] employ a different protected Hamiltonian and are not covered by no-go arguments

derived here for isolated fixed EO/SOEO blocks. Novelty relative to the full composite-pulse and dynamically corrected gate literature has **not** been established.

Throughout, labels mean: **ANALYTICAL** (derivation under explicitly stated assumptions), **NUMERICAL** (reproducible matrix verification at finite precision), **MODEL-ONLY** (estimated rates under specified synthetic noise), and **OPEN** (unrealized or unverified target). We do not equate a Monte Carlo model with experimental gate fidelity.

2. Spin encoding, pulse convention, and robustness criterion

For spin-$1/2$ particles, let $P_{ij}^{(s)} = (I - \mathrm{SWAP}_{ij})/2$. A positive exchange pulse of area a is

$$X_{ij}(a) = \exp\!\big(i\pi a P_{ij}^{(s)}\big), \qquad a > 0.$$

We distinguish ideal pulse periodicity $X_{ij}(a+2) = X_{ij}(a)$ from *noisy* periodicity: with fractional gain miscalibration the two physical implementations have different derivatives. In the strict model at most one physical bond is active per time step. Additional simultaneous-coupling schemes are always labeled separately.

For the full physical Hilbert-space propagator $U(\delta)$, with a gain δ_b shared by all pulses acting on bond b, let E be the isometric embedding of all code columns (including both EO gauge sectors) and $P = EE^\dagger$. A physically meaningful first-order condition for target encoded gate U_T is

$$U(0)E = e^{i\phi} E U_T, \qquad G_b E = c_b E, \quad G_b = -iU(0)^\dagger \partial_{\delta_b} U\big|_0.$$

Here G_b is Hermitian, c_b is real, and the second condition includes both **non-scalar logical/gauge errors** and **leakage**. The equivalent decomposition is $E^\dagger G_b E = c_b I$ and $(I - EE^\dagger) G_b E = 0$. For chronological pulses, up to the sign convention of the pulse phase,

$$G_b = \pi \sum_{k:b_k=b} a_k V_{k-1}^\dagger P_b^{(s)} V_{k-1},$$

where V_{k-1} is the ideal prefix before pulse k. This definition does **not** permit a favorable average over gauge states to hide harmful gauge dependence.

The main noise proxy is $a_k^{\mathrm{actual}} = a_k(1 + \delta_{b_k} + \eta_k)$, with $\delta_b \sim N(0, \sigma_{\mathrm{pb}}^2)$ independent across bonds but fixed throughout a word, and $\eta_k \sim N(0, \sigma_{\mathrm{iid}}^2)$ independent for every pulse. Unless stated, idle, shuttling, readout, reset, crosstalk, and context errors are omitted.

3. Spin-sector revival: a protected *identity* and probabilistic detector

Consider four SOEO data spins with total $\mathbf{S}_D = \sum_{i=1}^4 \mathbf{S}_i$ and an ancilla singlet $|S_{ab}\rangle$. Spin b is a spectator during a star pulse in which **four** bonds to spin a are activated simultaneously:

$$H_\star = J \sum_{i=1}^4 \mathbf{S}_i \cdot \mathbf{S}_a = J \mathbf{S}_D \cdot \mathbf{S}_a.$$

For every encoded $S_D = 0$ state, $H_\star P_0 = 0$ and thus $U_\star(t)P_0 = P_0$ for any t: this is a dark subspace, not a useful logical rotation. For data total-spin sector $S = 0, 1, 2$, the ancilla-singlet return Kraus amplitude is

$$A_S(t) = \frac{(S+1)e^{-iJSt/2} + Se^{iJ(S+1)t/2}}{2S+1}.$$

At the revival time $JT = 2\pi$,

$$K_s = P_0 - \tfrac{1}{3}P_1 + \tfrac{1}{5}P_2, \qquad E_t = I - K_s^\dagger K_s = \tfrac{8}{9}P_1 + \tfrac{24}{25}P_2.$$

Thus an ideal single-spin Pauli error from a valid four-spin singlet is flagged with probability $8/9$ per check. After n independent ideal repeats using new singlet pairs, the miss probability for a persistent $S_D = 1$ error is 9^{-n}. This is **not** a one-shot projective P_0 measurement, nor a proof of finite-round distance-three QEC.

For independent **static** gain biases $J_i = J(1+\delta_i)$, the perturbation $V = J\sum_i \delta_i \mathbf{S}_i \cdot \mathbf{S}_a$ connects an initial $S_D = 0, F = 1/2, E = 0$ state at first order only to the $S_D = 1, F = 1/2, E = -J$ sector. The Dyson term contains

$$\partial_{\delta_i} U(T)P_0 \propto \int_0^{2\pi/J} e^{iJt}\, dt = 0.$$

Hence every individual star-bond first derivative vanishes on the encoded input, and the single-operation state infidelity begins at $O(\|\delta\|^4)$. **This cancellation applies to the dark identity and only to the stated time-independent isotropic-gain model.** An $O(\delta^2)$ effective *logical* Hamiltonian remains for unequal gains: letting $\mathbf{D} = \sum_i \delta_i \mathbf{S}_i$,

$$H_{\text{eff}}^{(2)} = \frac{J}{4} P_0 \mathbf{D}^2 P_0.$$

The associated coherent logical rotation can accumulate over cycles. The research notes previously reported a model-level accepted-state infidelity near 5×10^{-7} per star probe at $\sigma_{\text{pb}} = 0.015$ but near 5×10^{-3} after 100 cycles with persistent gain biases. These quantities are **not** decoded logical QEC failure rates. The star also violates the strict one-bond-at-a-time hardware contract; naive serial Trotterization does not preserve its finite-pulse guarantees.

4. Nontrivial encoded gate mechanisms: what survives and what fails

A smaller two-bond simultaneous star $H_{2J} = J(\mathbf{S}_1 + \mathbf{S}_2) \cdot \mathbf{S}_a$ has an exact revival at $T = 4\pi/(3J)$:

$$U_{2J}(T) = P_{12}^{(s)} + e^{-2\pi i/3} P_{12}^{(t)}.$$

Because pair singlet/triplet distinguishes the customary encoded $|0_L\rangle, |1_L\rangle$ in EO and SOEO, this acts as a nontrivial encoded $R_{Z_L}(-2\pi/3)$. It uses two simultaneous bonds and does **not** cancel both independent gain errors. A three-segment $2J$ echo with two interleaved physical SWAPs suppresses the *differential* first-order error under the ideal-SWAP assumption, but the common gain and SWAP bond errors remain. A direct one-bond phase pulse outperformed these mechanisms in the source model at $\sigma_{\text{pb}} = 1.5\%$.

Perturbative SOEO–SOEO and EO–SOEO two-block couplers were also derived from gap-protected virtual transitions, with $J_{ZZ} \propto g_1 g_2/\Delta$. They are not exact native CZ gates; their useful phase is itself first-order sensitive to fractional g_1, g_2, and Δ variations, and the tested parameters led to long operation times. EO-to-SOEO conversion through a total-spin-zero projection was found as a heralded isometry of ideal success probability $1/2$ given a known EO gauge and one polarized added spin, not a deterministic coding conversion. These constructions are retained as *candidate mechanisms*, not recommended gate frontiers.

5. Strictly serial gain-robust local Clifford primitives

Composite words alternate two noncommuting physical bonds. The 11-pulse Hadamard was optimized for lower total positive exchange area relative to an earlier 9-pulse candidate. Full precision chronological angles are included in `sequences/`. For EO, the two bonds are J_{12}, J_{23}; for SOEO, the numerically verified mapping is J_{12}, J_{13}. The two encoded logical axes have the same 2×2 representation, so a shared set of pulse angles implements the same logical target; the full-spin calculations verify gauge independence and absence of nominal leakage.

Target	Word	Pulse count	Total area $\sum a_k$	EO target residual	SOEO target residual
H_L	H11	11	8.308609	7.10×10^{-14}	5.03×10^{-14}
$S_L = \mathrm{diag}(1, i)$	S13	13	8.858912	1.20×10^{-14}	8.49×10^{-15}
$Z_L = \mathrm{diag}(1, -1)$	Z13	13	7.838015	9.24×10^{-14}	6.53×10^{-14}

All active-bond first-order non-scalar tangent norms for the listed EO words are below 3×10^{-13} in the supplied double-precision verifier, with nominal leakage near 10^{-15} or below. The supplementary archive also contains an 11-pulse S candidate with *larger* area (10.324126π). The values constitute **numerically exact**, not symbolic transcendental, pulse solutions; no globally minimal length or duration has been proven.

The prescribed synthetic noise model with $\sigma_{\mathrm{pb}} = 0.015$ and $\sigma_{\mathrm{iid}} = 0.0006$ gave the following averages (the fidelity definition follows the source package's encoded/gauge treatment):

Local gate	Native short word, $1 - F$	Selected robust word, $1 - F$	Approx. ratio
H_L: 3P vs 11P	9.37×10^{-4}	1.36×10^{-5}	$69\times$
S_L: 1P vs 13P	9.95×10^{-4}	1.02×10^{-5}	$97\times$
Z_L: 1P vs 13P	4.43×10^{-4}	9.05×10^{-6}	$49\times$

These ratios are *not* hardware advantages. The 11P H has 8.31π total area versus approximately 2π for the 3P reference and eight additional switch events. A modeled area-dependent penalty exceeding roughly 1.46×10^{-4} per additional normalized area unit could reverse the benefit under the original source assumptions; this crossover is illustrative, not measured. In syndrome extraction a conjugated two-data XX instrument would cost $47 + 4 \times 3 = 59$ pulses with basic H, but $47 + 4 \times 11 = 91$ pulses with robust H, *conditional* on the unavailable archived 47P physical source word. The full 59P/91P instruments were **not** independently revalidated.

6. Six-spin encoded entangler: exact 43P to 41P rewrite

For two EO blocks $A = (1, 2, 3)$ and $B = (4, 5, 6)$ in a linear chain, an archived 43-pulse source word was found despite its misleading CZ25 filename. The encoded target in its explicit basis is a **locally rotated** controlled reflection

$$U_{\text{rotCZ}} = I_4 - 2|vv\rangle\langle vv|, \qquad |v\rangle = (\sqrt{3}|0_L\rangle + |1_L\rangle)/2,$$

with identity action on both gauge qubits. It is locally equivalent to a literal computational-basis CZ; the provided CSV is **not** the latter without a local-frame conversion. Two exact physical rewrite identities merge disjoint-bond-adjacent pulses, and one merger replaces a nominal pulse of area 2.5 by 0.5 using $X_b(a + 2) = X_b(a)$. The resulting **41P** word is identical to the 43P word on all 64 physical states to Frobenius norm 1.68×10^{-15}, with encoded leakage 1.75×10^{-15}. The change of angle branch reduces the derivative with respect to the associated physical bond, although it leaves the ideal gate unchanged.

We define a same-convention, normalized squared tangent measure

$$h_b = \frac{\|E^\dagger G_b E - c_b I\|_F^2 + \|(I - EE^\dagger)G_b E\|_F^2}{16}, \qquad c_b = \frac{\text{Tr}(E^\dagger G_b E)}{16}.$$

Bond	43P h_b	41P h_b
J_{12}	63.5527	39.1871
J_{23}	24.1692	24.1692
J_{34}, interblock	33.3549	33.3549
J_{45}	18.3557	18.3557
J_{56}	9.4908	9.4908
Total	**148.9232**	**124.5576**

The total coefficient decreases by 16.4%, but **no active bond** meets the full zero-tangent condition. In particular the cross-bond response is unchanged. A paired 850-trial model comparison at $\sigma_{\text{pb}} = 0.015$ and $\sigma_{\text{iid}} = 0.0006$ yielded phase-optimized encoded operator distances $D_{43} = 0.03380$ and $D_{41} = 0.02817$; this D is **not** Haar-average gate infidelity. The 41P result is neither a shortest CZ nor a fully robust entangler; existing 25P/27P gate baselines in the wider project were not recertified in this archive.

7. Necessary constraints on fully robust encoded entanglers

Single-use-bond theorem (ANALYTICAL). If a physical bond occurs only once in a word and its independent gain derivative must be a scalar phase with no first-order leakage on the transported input code, its conjugated singlet projector must restrict to eigenvalue 0 or 1 on that transported code. Therefore that pulse acts as an overall scalar on the code and is removable from the *ideal encoded action*. If every cross-block bond occurs at most once and each satisfies exact individual-bond robustness, the remaining internal exchange operations cannot generate entanglement. Thus a nontrivial all-bond-first-order-robust entangler needs at least one repeated nontrivial cross bond, or additional hardware resources/changed code assumptions. This is **not** a global CZ no-go theorem.

Single-cross-pulse obstruction (ANALYTICAL + NUMERICAL). For $U = L_{\text{post}} X_{ij}(a) L_{\text{pre}}$, with $L_{\text{pre/post}}$ restricted to inside fixed EO/SOEO blocks, the leakage norm is

$$\|QUP\|_F = 2|\sin(\pi a/2)|\, \|QP_{ij}^{(s)}P\|_F.$$

The rightmost factor is independent of which physical spin in a block is used:

Block pair	Physical spins	Encoded rank	$\|QP_{ij}^{(s)}P\|_F$
EO–EO	6	16	$\sqrt{56/27} \approx 1.44016$
EO–SOEO	7	8	$\sqrt{3/2} \approx 1.22474$
SOEO–SOEO	8	4	$\sqrt{3/4} \approx 0.86603$

A single nontrivial isolated cross-block pulse consequently leaks from the fixed code. For the pairs involving a singlet SOEO block, the projected cross-block singlet operator is only $PP_{ij}^{(s)}P = P/4$. This does **not** contradict gate constructions with always-on intra-block Hamiltonians, encoding changes, measurement-assisted control, or more than one cross pulse.

Local-dressing theorem. If $U = L_{\text{post}} E L_{\text{pre}}$ and a gain parameter acts only in the noisy entangling core E, the harmful derivative is merely unitarily conjugated by L_{pre}. Consequently putting individually robust H/S gates around an unchanged CZ **cannot** cancel its interblock gain tangent. The core itself, or the physical resources, must change.

8. Implications for QEC and priorities

The source hybrid SOEO paper measures physical-spin $XXXX/ZZZZ$ within a *single* four-spin encoding using addressable LD ancillas. Our strict EO objective instead concerns encoded logical parity instruments and the actual physical measurement branches; the unitary equality $H_L Z_L H_L = X_L$ is insufficient to certify the entire noisy instrument. Any candidate must be tested on the full encoded logical$\times$gauge space, the correct ancilla preparation/readout channel, leakage sectors and single-fault hook propagation, and persistent bond gains over multiple rounds. Conditional postselection fidelity, a one-shot parity readout probability, and a decoded logical error per real-time cycle are different metrics.

The order of priority is: (i) obtain and recertify the original **shortest geometry-matched** six-spin CZ source CSVs; (ii) synthesize a short core with repeated cross-bond interfaces, optimizing all full-space tangent matrices rather than one scalar; (iii) compare the same core in a physical QEC instrument under static gain, IID pulse jitter, idle, readout/reset, and leakage-aware decoding; (iv) evaluate whether robust local frames improve total syndrome cost after pulse duration and switching overhead. Until those tasks are complete, the project's strongest result is a set of **local robust primitives and an informative restricted entangler obstruction**, not a practical QEC threshold or fully gain-robust CZ.

9. Reproducibility and limitations

The main local and six-spin code verifiers were rerun on 1 November 2026 using NumPy, with single-threaded BLAS. For local gates, `code/verify_all_physical.py` tests both EO gauge sectors in the 8D physical space and the 16D SOEO singlet subspace, and `code/replay_benchmark.py` reruns a 3,000-trial stochastic benchmark. For the entangler, an independent physical SWAP/permutation implementation verifies the 41P/43P ideal equivalence, `code/cz43_audit.py` returns the full bond tangent decomposition, and `code/single_bridge_no_go.py` tests all nine EO–EO interface choices. The accompanying archives include original chronological CSVs and provenance. They may include additional exploratory scripts that retain working-session paths; such scripts require adaptation before external reproduction. No hardware was controlled, no experimental samples were obtained, and no multiperiod stabilizer decoding was run.

The calculations should not be submitted as a novelty-centered paper without (a) a systematic quantitative comparison against [2,3] and later dynamically corrected EO literature; (b) independent analytic or high-precision certification of composite words; (c) a fair physical noise/time budget; and (d) a new entangling-gate or end-to-end QEC gain. This report is a transparent technical research artifact.

References

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